

CoRoT-1b

R-2589

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_170109 · created 2026-08-27T23:09:49+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457762.8668 +/- 0.0034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.012
Transit depth (Rp/Rs)^2	2.05 +/- 0.33 [%]
Orbital Inclination (inc)	83.3 +/- 2.62
Airmass coefficient 1 (a1)	1.028 +/- 0.065
Airmass coefficient 2 (a2)	-0.024 +/- 0.011
Scatter in the residuals of the lightcurve fit is	6.25 %
Best Comparison Star	#3 - [142, 275]
Optimal Aperture	3.29
Optimal Annulus	11.45
Transit Duration (day)	0.097 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.143 ± 0.012 vs 0.1388 (deviation 0.18σ)
Duration fit vs published	0.097 d vs 0.1046 d (deviation 0.27σ)
Mid-time O–C	-4.9 min
Depth SNR	6.3
Residual scatter	6.25 %

Light curve

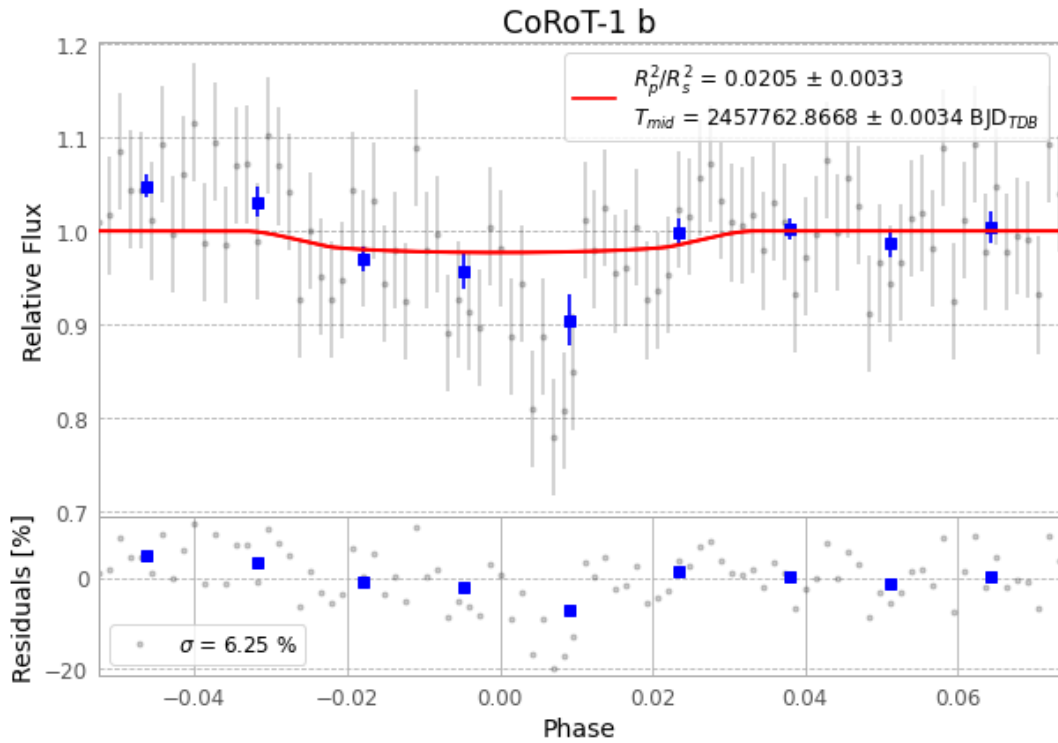


Figure 1 — FinalLightCurve_CoRoT-1 b_2017-01-08.png (SHA-256 9edd67ae06a7ccd3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [171, 287], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c38f841def1b219d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

92 FITS frames (60 MB), CoRoT-1170109064512.FITS → CoRoT-1170109111812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-170109.log (b26f7c05f29f...), FinalLightCurve_CoRoT-1 b_2017-01-08.png (9edd67ae06a7...), FinalLightCurve_CoRoT-1 b_2017-01-08.csv (7fdc23227aa1...)

Compute resources

Wall time 133.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-27 23:09Z.